GRIT Interview Scheduling: Write-up

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I designed the feature around one idea: every student has **one interview journey per badge**, and that journey is always in exactly **one state**. Every screen (student, interviewer, interview team) reads the same state, so all three roles always see the same truth. All rules live in the backend API, and the database enforces the ones that matter when two people act at the same moment.

1. States

State	What the student sees	Moves to	What moves it
not_eligible	Skills left for this badge	eligible	The last required skill is cleared (the skills service recomputes eligibility)
eligible	"You're interview-ready" + Book	booked	Student books a slot 6 h to 14 days ahead
booked	Ticket: date, time, interviewer, join link 10 min before	eligible · awaiting_result	Student or interviewer cancels (an interviewer cancel gives the student priority to rebook) · the slot's start time passes
awaiting_result	"Happening now", then "Your result is on its way"	passed · retake_later · failed · locked	Interviewer records Pass · Retake later · Fail or No-show · that failure was attempt 3
retake_later	"One more round needed", bookable from a date	eligible	The retake date arrives (not counted as an attempt)
failed	"You didn't clear it this time", rebook date, attempts left	eligible	Cooldown ends: 14 days after a Fail, 7 days after a No-show
locked	"Your attempts are used up"	—	End state after 3 failed attempts
passed	"Badge unlocked", +500 Miles, Qualiflyer leaderboard	under_review	The team raises a malpractice flag
under_review	"Your badge is under review" (paused on leaderboard)	passed · nullified	The team dismisses the flag · upholds it
nullified	"Your badge has been withdrawn", Miles removed	—	End state

Two moves depend only on time (booked → awaiting_result, and a cooldown ending back in eligible). I derive these when the data is read instead of running a job, so the screen is never out of date. An interviewer can edit a result for 2 hours; I restore the journey from a saved snapshot and reverse the Miles by the difference. Every move writes an audit event that can't be edited or deleted.

2. Edge cases and questions

Cases I handle

- **Two students click the same slot:** the database allows only one confirmed booking per slot (and one per journey). The second student sees "That slot was just taken. Pick another one."
- **Booking rules:** slots from 6 hours to 14 days ahead; cancel or reschedule until 12 hours before; at most 2 reschedules.
- **Interviewer can't make it:** they cancel with a reason and the student goes back to eligible with priority, or the team reassigns the booking to another free interviewer at the same time.
- **Overlapping availability:** the database rejects overlapping windows for the same interviewer; a window that has a booking can't be removed.
- **Bad or missing results:** a result can only be recorded after the interview starts, all 4 rubric areas must be scored 1–4, Fail and Retake need at least 15 characters of feedback, and a retake date must be in the future. Interviewers get reminders at 2 h and 20 h; the team sees it as overdue at 24 h.
- **No-show:** counts as an attempt with a 7-day cooldown. Retake later never counts.
- **Malpractice found after a Pass:** the badge goes under review first, so nothing is removed by mistake. Upholding removes the badge and its 500 Miles; dismissing restores it.
- **Double clicks and retries:** booking and result requests carry an idempotency key, so a repeated click doesn't create a second booking.
- **Time zones:** stored in UTC, always shown in IST.
- **Roles:** students self-register and are always created as students; interviewers and admins are added by the team. The API checks the role on every request.

Assumptions

- Interviews are 30 minutes, online, on a fixed Google Meet room per interviewer.
- 3 attempts per badge; a 14-day cooldown after a Fail.
- Eligibility comes from the existing skills service; students can't declare it themselves.
- An interviewer only interviews for the badges they are cleared for.

Open questions

- Who can unlock a locked journey, and after how long?
- Should a student be able to appeal a withdrawn badge in the app, or only by contacting the team?
- Do interviewers need a daily interview cap, or is their availability enough?

3. Data model

Postgres (Supabase), 19 tables. The journey row is the state machine; bookings, outcomes and awards hang off it.

Table	Key fields	Rules in the schema
profiles	id (= auth user), role, full_name, email	role is student, interviewer or admin
students / **interviewers**	campus, batch, miles / campus, meet_link, active	miles ≥ 0
badges, **skills**, **badge_skill_requirements**	track, level 1–5; minimum skill level per badge	one badge per track and level
student_skill_levels	student, skill, level, status	filled by the skills service
availability_windows	interviewer, starts_at, ends_at, deleted_at	no overlapping live windows (exclusion constraint)
slots	window, start_at, end_at, status (open / booked / blocked / cancelled)	unique per interviewer and start time
interview_journeys	student, badge, state, attempts_used, next_bookable_at, active_booking_id, priority	one per student and badge; attempts 0–3
bookings	journey, slot, status, meeting_link, reschedule_count, cancel_reason	one confirmed booking per slot and per journey; reschedules 0–2
interview_outcomes	booking, result, rubric, feedback_to_student, internal_notes, retake_after, prev_snapshot, locked_at	one per booking; feedback ≥ 15 characters for Fail and Retake
badge_awards	student, badge, outcome, miles_awarded, status	one live award per student and badge
malpractice_flags	award, raised_by, evidence_url, status, decision_note	open / upheld / dismissed
journey_events	journey, from_state, to_state, actor, reason	append-only audit log
notifications	user, template, title, body, tone, read_at	in-app today; email and WhatsApp reserved

A view (`v_journey_effective`) applies the time-based moves when data is read, and another (`v_leaderboard`) ranks Qualiflyers and Pre-Qualiflyers by Miles. Row-level security limits each role to its own rows.